

JARRETT D. PHILLIPS, Ph.D.
School of Computer Science
Department of Integrative Biology
University of Guelph
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GitHub: [jphill01](#)

RESEARCH AREAS

Computational statistics · Data science · Machine learning · Statistical modelling · Biodiversity informatics · eDNA and DNA barcoding

ACADEMIC APPOINTMENTS

Adjunct Professor , School of Computer Science University of Guelph	2023–Present
Affiliated Faculty , One Health Institute University of Guelph	2026

EDUCATION

Ph.D. in Computational Sciences , University of Guelph Co-Advisors: Dr. Daniel Gillis and Dr. Robert Hanner Advisory Committee Members: Dr. Deborah Stacey and Dr. Graham Taylor Dissertation: A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation	2022
Master of Bioinformatics , University of Guelph Co-Advisors: Dr. Robert Hanner and Dr. Daniel Ashlock Major Paper: Assessing DNA Barcode Haplotype Sampling Diversity in the Ray-finned Fishes (Chordata: Actinopterygii)	2014
BSc. (Hons.) in Biological Science, University of Guelph Coursework in bioinformatics, ecology, evolutionary biology, comparative animal physiology, genetics, mathematics, and statistics	2013

RESEARCH EXPERIENCE

Postdoctoral Fellow University of Guelph GBADs Informatics Team Stacey Lab, School of Computer Science Supervisor: Dr. Deborah Stacey	2023–Present
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Postdoctoral Fellow 2023–Present
University of Guelph
Gillis Lab, School of Computer Science
Hanner Lab, Department of Integrative Biology
Supervisors: Drs. Daniel Gillis and Robert Hanner

Postdoctoral Fellow 2022
University of Guelph
Hanner Lab, Department of Integrative Biology
Supervisor: Dr. Robert Hanner

RESEARCH FUNDING

Completed

6. Food From Thought Advancing Research Impact (ARIF) Fund – Livestock Innovation Grant 2024
University of Guelph \$40000 CAD
A Modular Decision- and Policy-Making Tool for Global Livestock Disease Burden Assessment
Role: Co-applicant

5. Food From Thought Advancing Research Impact (ARIF) Fund 2023
University of Guelph \$30000 CAD
Forecasting Seafood Fraud Occurrence in the Supply Chain Within Major Canadian Cities Using Bayesian Hierarchical Binary Logistic Time-Series Regression Modeling
Role: HQP

4. SoCS Travel Grant 2019
University of Guelph \$1000 CAD

3. Arthur D. Latornell Graduate Travel Grant 2019
University of Guelph \$500 CAD

2. CPES Graduate Dean's Scholarship 2017
University of Guelph \$3500 CAD

1. CPES Graduate Excellence Entrance (GEE) Scholarship 2016
University of Guelph \$30000 CAD

Submitted

3. NSERC Discovery Grant 2026
University of Guelph Amount \$TBD CAD
Computational Statistical Methods for Specimen Sampling Sufficiency and Species Genetic Diversity Assessment
Role: Principal Investigator

2. NSERC Alliance Society Grant 2026
University of Guelph \$2.5 million CAD
AI-Driven Biodiversity Restoration: Scalable Solutions for Ecosystem Monitoring and Conservation
Role: Collaborator

1. NSERC CREATE Grant 2026
University of Guelph \$9.9 million CAD
AI-Driven Training of Next-Generation Conservation Scientists
Role: Collaborator

To be Submitted

3. Advancing Research Impact Fund: “Quick Wins” Knowledge Mobilization Fund 2026
University of Guelph \$20000 CAD
GBADs ModelBuilder as a Tool to Teach Livestock Population Dynamics and Disease Burden Risk Estimation
Role: Co-applicant

SOFTWARE AND COMPUTATIONAL TOOLS

95,000+ software downloads through the Comprehensive R Archive Network (CRAN)

- 6. **Seafood Fraud Modelling App** · Python web application
- 5. **GBADs ModelBuilder** · R Shiny web application
- 4. **RulesTools** · R package · >3000 downloads · DOI: [10.32614/CRAN.package.RulesTools](https://doi.org/10.32614/CRAN.package.RulesTools)
- 3. **Frequentist/Bayesian DNA barcode gap coalescent-informed estimators** · R scripts
- 2. **VLF (Very Low Frequency)** · R package · >40000 downloads · DOI: [10.32614/CRAN.package.VLF](https://doi.org/10.32614/CRAN.package.VLF)
- 1. **HACSim (Haplotype Accumulation Curve Simulator)** · R package · R Shiny web application · >50000 downloads · DOI: [10.32614/CRAN.package.HACSim](https://doi.org/10.32614/CRAN.package.HACSim)

STUDENT SUPERVISION AND MENTORSHIP

Mentored/co-supervised **18 undergraduate and graduate students** across computer science, bioinformatics, statistics, and biology.

- * Indicates students under my direct mentorship/supervision
- ** Indicates students under my co-mentorship/co-supervision

Undergraduate

***Brendan Carl Rosario** 2025
Undergraduate Student Volunteer · Supervised Machine Learning for eDNA spatiotemporal sampling
***Richard Cui** 2025
Summer Undergraduate Research Assistant (URA) · Dynamical modelling of seafood fraud in the supply chain

*Nikolett Toth	2024-2025
CIS*4900/4910 · Mining association rules for eDNA spatiotemporal sampling	
*Nikolett Toth	2024
Summer Undergraduate Research Assistant (URA) · Mining association rules for eDNA spatiotemporal sampling	
*Fynn De Vuono-Fraser	2024
CIS*4900/4910 · Bayesian modelling of seafood fraud in the Canadian supply chain	
*Zaid Al-Gayyali	2023
Summer Undergraduate Research Assistant (URA) · Seafood Fraud Visualization Tool Shiny app	
*Fynn De Vuono-Fraser	2023
STAT*4600 · Bayesian modelling of seafood fraud in the Canadian supply chain	
*Navdeep Singh	2021
CIS*4900 · HACSim R Shiny web application	
*Scarlett Bootsma	2020–2021
CIS*4900/4910 · HACSim simulation study	
**Maya Persram	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
**Ashley Chen	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
**Olivia Friesen Kroeker	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
*Steven French	2018
CIS*4900/4910 · HACSim R package	
**Julia Harvie	2018–2019
MCB*4500/4510 · Data mining GenBank and BOLD	

Graduate

**Nathan Zeinstra	2024
IBIO*6070 · Bayesian habitat occupancy modelling of sea lamprey using eDNA	
**Amina Asif	2022
BINF*6999 · DNA barcode gap analysis of Canadian disease vectors and agricultural pests	
**Danielle St. Jean (withdrawn)	2018–2019
MSc. thesis (Mathematics) · DNA barcode sequence classification with machine learning	
**Christina Fragel	2018–2019
BINF*6999 · DNA barcode sequence classification with machine learning	
**Jiaojia (Paula) Yu	2018–2019
BINF*6999 · MDMAPR R Shiny app	
**Ankita Bhanderi	2018
BINF*6999 · Data mining GenBank and BOLD	

TEACHING EXPERIENCE

Course Instructor University of Guelph CIS*1910 – Discrete Structures in Computing I	2025
Graduate Teaching Assistant (GTA) University of Guelph CIS*3130 – System Modelling and Simulation CIS*1910 – Discrete Structures in Computing I CIS*2460 – Modelling of Computer Systems	2016–2020 2020 2017 2016–2019

ACADEMIC SERVICE

6. Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference 2026 University of Guelph Conference organizer and volunteer	
5. 39th Canadian Artificial Intelligence Conference Simon Fraser University Reviewer for Long and Short Papers Track	2026
4. GBADs Informatics Technical Workshop University of Liverpool Workshop organizer	2023
3. Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference University of Guelph Conference organizer and volunteer	2023
2. School of Computer Science (SoCS) Faculty Search Committee University of Guelph Member of hiring panel for Associate Professor in Cybersecurity	2018
1. School of Computer Science (SoCS) Faculty Search Committee University of Guelph Member of hiring panel for 2-year Contractually Limited Assistant Professor in Cybersecurity	2017–2018

CONFERENCE PRESENTATION AND PROCEEDINGS

* Indicates students under my direct mentorship/supervision

16. The VII AMMCS International Conference (student oral presentation) Wilfrid Laurier University, Canada · Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain	2026
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15. **Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference** (oral presentation) 2026
University of Guelph, Canada
Phillips, J.D. · Association rule mining for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling
14. **Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference** (poster presentation) 2026
University of Guelph, Canada
*Toth, N., Antonie, M.L. and **Phillips, J.D.** · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets
13. **39th Canadian Conference on Artificial Intelligence** (short paper and student poster) 2026
Simon Fraser University, Canada
*Toth, N., Antonie, M.L. and **Phillips, J.D.** · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets.
12. **SoCS Undergraduate Summer Project Show and Tell** (student oral presentation) 2025
University of Guelph, Canada
*Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** · The Importance in Design as a Computer Scientist
11. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2024
University of Guelph, Canada
*Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets
10. **CBS Undergraduate Poster Session** (student poster presentation) 2024
University of Guelph, Canada
*Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets
9. **9th International Barcode of Life Conference** (poster presentation) 2024
Estação das Docas, Belem, Brazil
Phillips, J.D. · A Measure of the DNA Barcode Gap for Applied and Basic Research (*not attended*)
8. **GBADs Informatics Technical Workshop** (oral presentation) 2023
University of Liverpool, England
Phillips, J.D. · The GBADs R Package (and Why We Need It!)
7. **CEPS Student Research Day** (student poster presentation) 2023
University of Guelph, Canada
*De Vuono-Fraser, F.A. and **Phillips, J.D.** · Estimating Seafood Mislabelling Rates in Canada Using Bayesian Modelling

6. **Pathway to Increase Standards and Competency of eDNA Surveys (PISCeS) International Conference** (poster presentation) 2023
University of Guelph, Canada
Phillips, J.D. · Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes
5. **8th International Barcode of Life Conference** (oral presentation) 2019
NTNU University Museum and Norwegian Biodiversity Information Centre, Norway
Phillips, J.D. · HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness
4. **Guelph BioMathematics and Statistics (BioM&S) Symposium Artificial Intelligence and Machine Learning in Biology** (attended) 2019
University of Guelph, Canada
3. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2018
University of Guelph, Canada
*French, S.H. and **Phillips, J.D.** · Estimating Sampling Size Using Haplotype Accumulation Curves and Semiparametric Models
2. **7th International Barcode of Life Conference** (oral presentation). 2017
University of Johannesburg, South Africa
Phillips, J.D. · Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough?
1. **6th International Barcode of Life Conference** (poster presentation) 2015
University of Guelph, Canada
Phillips, J.D. · An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii).

REFEREED WORK

Publication record: 8 peer-reviewed journal articles (6 first-author, 2 co-author) and 2 book chapters (1 first-author, 1 co-author), plus 3 first-author manuscripts in preprint or under review. Senior (last) author on 2 student-led manuscripts

* Indicates students under my direct mentorship/supervision

** Indicates students under my co-mentorship/co-supervision

Journal Articles

Published or Accepted

9. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2026 Swimming in deep uncertainty: How proper statistical modelling can help expose seafood product mislabeling. *CHANCE*, 3.
DOI: 10.1080/09332480.2026.2725526

8. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2026). Statistical modelling of seafood fraud highlights uncertainties in products from Metro Vancouver, British Columbia, Canada: Revisiting Hu *et al.* (2018). *Journal of Food Science*, **91**: e71201. DOI: [10.1111/1750-3841.71201](https://doi.org/10.1111/1750-3841.71201)
7. Raymond, K., Sobkowich, K.E., **Phillips, J.D.**, Nguyen, L., McKechnie, I., Mohideen, R.N., Fitzjohn, W., Szurkowski, M., Davidson, J., Rushton, J., Stacey, D.A., and Bernardo T.M. (2024). GBADs informatics strategy: User-centric tools, data quality, and model interoperability. *WOAH Scientific and Technical Review*, **43**: 96-107. DOI: [10.20506/rst.43.3522](https://doi.org/10.20506/rst.43.3522).
6. **Phillips, J.D.**, Athey, T.B.T., Hanner, R.H., and McNicholas, P.D. (2023). VLF: An R package for the analysis of very low frequency variants in DNA sequences. *Biodiversity Data Journal*, e96480. DOI: [10.3897/BDJ.11.e96480](https://doi.org/10.3897/BDJ.11.e96480).
5. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2022). Lack of statistical rigor in DNA barcoding likely invalidates the presence of a true species' barcode gap. *Frontiers in Ecology and Evolution*, **10**: 859099. DOI: [10.3389/fevo.2022.859099](https://doi.org/10.3389/fevo.2022.859099).
4. D'Ercole, J., Dincă, V., Opler, P.A., Kondla, N.G., Schmidt, C.B., **Phillips, J.D.**, Robbins, R., Burns, J.M., Miller, S.E., Grishin, N., Zakharov, E.V., deWaard, J.R., Ratnasingham, S., and Hebert, P.D.N. (2021). A DNA barcode library for the butterflies of North America. *PeerJ*, **9**: e11157. DOI: [10.7717/peerj.11157](https://doi.org/10.7717/peerj.11157).
3. **Phillips, J.D.**, *French, S.H., Hanner, R.H., and Gillis, D.J. (2020). HACSim: An R package to estimate intraspecific sample sizes for genetic diversity assessment using haplotype accumulation curves. *PeerJ Computer Science*, **6**(192): 1-37. DOI: [10.7717/peerj-cs.243](https://doi.org/10.7717/peerj-cs.243).
2. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2019). Incomplete estimates of genetic diversity within species: Implications for DNA barcoding. *Ecology and Evolution*, **9**(5): 2996-3010. DOI: [10.1002/ece3.4757](https://doi.org/10.1002/ece3.4757).
1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii). *DNA Barcodes*, **3**: 66-73. DOI: [10.1515/dna-2015-0008](https://doi.org/10.1515/dna-2015-0008).

Preprints

3. *Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. bioRxiv. DOI: [10.64898/2026.09.16.752056](https://doi.org/10.64898/2026.09.16.752056).
2. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. (2025). A Bayesian coalescent model of the DNA barcode gap. Authorea. DOI: [10.22541/au.174073683.37707806/v1](https://doi.org/10.22541/au.174073683.37707806/v1)
1. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2024). Statistical modelling of seafood fraud in the Canadian supply chain. bioRxiv. DOI: [10.1101/2024.02.05.578947](https://doi.org/10.1101/2024.02.05.578947).

Submitted or Under Revision

1. *Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. Submitted to *Oikos*.

To Be Submitted

2. Cui, R.C., Yodzis, M.P., and Phillips, J.D. Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain. Springer Nature.

1. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. A Bayesian-informed coalescent model of the DNA barcode gap. Targeted to *Methods in Ecology and Evolution*.

In preparation (selected)

3. *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** Seafood market product fraud erodes both consumer trust and resource sustainability within the global supply chain. Targeted to *Proceedings of the Royal Society B: Biological Sciences*.

2. Elliott, T.A., Diao, J., **Phillips, J.D.**, and Adamowicz, S.J. Benchmarking imputation methods for insect trait data across missingness mechanisms and trait types. Targeted to *Methods in Ecology and Evolution*.

1. D'Ercole, J., Dapporto, L., **Phillips, J.D.**, Dincă, V.E., Vila, R., Talavera, G., and Hebert, P.D.N. Macrogenetics of North American butterflies—The impact of Quaternary climatic fluctuations. Targeted to *PNAS*.

Book Chapters

Published or Accepted

2. **Phillips, J.D.**, Griswold, C.K., Young, R.G., Hubert, N., and Hanner, R.H. (2024). A Measure of the DNA Barcode Gap for Applied and Basic Research. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY.

URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_24

1. Hubert, N., **Phillips, J.D.**, and Hanner, R.H. (2024). Delimiting Species with Single-Locus DNA Sequences. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY. URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_3

Submitted or Under Revision

1. *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain. VII AMMCS Conference, Springer Nature.

Conference Proceedings

Published or Accepted

5. *Toth, N., Antonie, M.L., and **Phillips, J.D.** To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets. *39th Canadian Conference on Artificial Intelligence*, **318**: 1012-1019. Short paper and poster.
4. Morey, K., Loeza-Quintana, T., **Phillips, J.**, and Hanner R. (2023). Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes. Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference. Poster.
3. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2019). HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness. Scientific abstracts from the 8th International Barcode of Life Conference, Trondheim, Norway (ed. Torbjørn Ekrem), *Genome*, 62(6): 349-453. Oral presentation.
2. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2017). Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough? Scientific abstracts from the 7th International Barcode of Life Conference, Johannesburg, South Africa (ed. M. van der Bank), *Genome*, 60(11): 881-1019. Oral presentation.
1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific haplotype diversity in the ray-finned fishes (Chordata: Actinopterygii). Scientific abstracts from the 6th International Barcode of Life Conference, Guelph, ON., Canada (ed. S.J. Adamowicz), *Genome*, 58(5): 163-303. Poster.

NON-REFEREED WORK

Ph.D. Thesis

Phillips, J.D. (2022). A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation. University of Guelph Atrium. URL: <https://atrium.lib.uoguelph.ca/items/8addfcc5-f21c-4691-89b7-c4db051892eb>

* Indicates students under my direct mentorship/supervision

15. *Cui, R.C. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
14. *Toth, N. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
13. *Cui, R.C. and **Phillips, J.D.** (2025). The Importance in Design as a Computer Scientist. Student-contributed SoCS Show and Tell video URL: <https://github.com/RichaCui70/Dynamics-between-Seafood-Fraudsters-and-Buyers/blob/main/Summer%20Showcase%20-%20Richard%20Cui.mp4>

12. **Phillips, J.D.**, *De Vuono-Fraser, F.A., Gillis, D.J., and Hanner, R.H. (2024). Statistical modelling of seafood fraud. Whiteboard explainer video. URL: <https://www.youtube.com/watch?v=4jU8cU8M70U>
11. *Toth, N. and **Phillips, J.D.** (2024). eDNA Collection Gets a Tech Update. Student-contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2024/07/edna-collection-gets-tech-upgrade>
10. *Toth, N. and **Phillips, J.D.** (2024). Unravelling eDNA with Association Rule Mining. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/08/15/unravelling-edna-with-association-rule-mining/>
9. **Phillips, J.D.** (2024). Summer URA Position. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/02/07/summer-ura-position-2/>
8. **Phillips, J.D.** (2022). A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation. Ph.D. Thesis. URL: <https://atrium.lib.uoguelph.ca/items/8addfcc5-f21c-4691-89b7-c4db051892eb>
7. **Phillips, J.D.** (2022). Mind the Gap — The DNA Barcode Gap, That Is. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2022/08/mind-gap---dna-barcode-gap>
6. **Phillips, J.D.** (2020). Barcode Cracking. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2020/02/barcode-cracking>
5. **Phillips, J.D.** (2020). Protecting Biodiversity Through the Lens of Genetic Diversity. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2020/01/30/protecting-biodiversity-through-the-lens-of-genetic-diversity/>
4. **Phillips, J.D.** (2019). IBOL8 and the Midnight Sun. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2019/07/02/reflections-ibol8-and-the-midnight-sun/>
3. **Phillips, J.D.** (2017). The Big Five and IBOL7. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2017/12/06/reflections-the-big-five-and-ibol7/>
2. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding: Are current sampling levels enough? Contributed guest post to the DNA Barcoding Blog of Dr. Dirk Steinke. URL: <http://dna-barcoding.blogspot.com/2016/01/guest-post-sample-size-estimation-for.html>
1. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding of ray-finned fishes: Are current sampling levels enough? Contributed newsletter article to the Barcode Bulletin, 7(1). URL: <https://journal.lib.uoguelph.ca/index.php/ibol/issue/view/345>

ACADEMIC PEER REVIEW SERVICE

Ecology and Evolution, F1000 Research, Frontiers in Ecology and Evolution, Lifestyle Genomics, Mitochondrial DNA Part A, Molecular Ecology Resources, Molecular Biology Reports, Methods in Ecology and Evolution, Nature Communications

VOLUNTEER EXPERIENCE

CEPS Student Research Connections Networking Night

2025

University of Guelph

- Connected with undergraduate students for summer URA project recruitment

CIS*3750 Wireframing session

2021-2025

University of Guelph

- Graded students on mobile app prototypes for various community partners using Qualtrics surveys